

COMP6237 Data Mining

Lecture 13: Revision

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Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Exam Format

- MCQs + Computer-aided
 - Shoaib (40 marks) + Zhiwu (40 Marks) + Markus (20 Marks)
- My part (15 MCQs):
 - MCQs 1-12 : 2 marks each (easy/normal), one single answer each
 - MCQs 13-15: 5 or 6 marks each (hard)
 - Each MCQ contains 2 or 3 subquestions
 - Each subquestion has one single answer

MCQ Examples

More examples: https://comp6237.ecs.soton.ac.uk/lectures/pdf/13-Revision-MCQ_examples_ZH.pdf

Question 3: How does the DBSCAN algorithm handle outliers in the dataset?

- A. Assigns outliers to the nearest cluster
- B. Removes outliers from the dataset
- C. Labels outliers as noise
- D. Forms separate clusters for outliers

Question 4: Which clustering algorithm is particularly effective for identifying clusters with varying shapes and densities?

- A. K-means
- B. DBSCAN
- C. Hierarchical clustering
- D. None of above

MCQ Examples

Question 5:

	Feature 1	Feature 2	Feature 3
Instance1	xx	xx	xx
Instance2	xx	xx	xx
Instance3	xx	xx	xx

Assume user A wanted to compute Euclidean distance for each pair of instances. Which option is correct?

- A. ...
- B. ...
- C. ...
- D. None of the above

MCQ Examples

Question 5: Given the 1D data points {1, 2, 3, 4, 6, 7, 8, 9, 10}, if k-means clustering is run with $k=2$ and initial cluster centers at $C1=2$ and $C2=8$ using Manhattan distance.

S1: Which points will be in the cluster with center 8 in the first iteration?

- A. 4
- B. 5
- C. 6
- D. 7

S2: Which are the two new centers:

A...

B....

C...

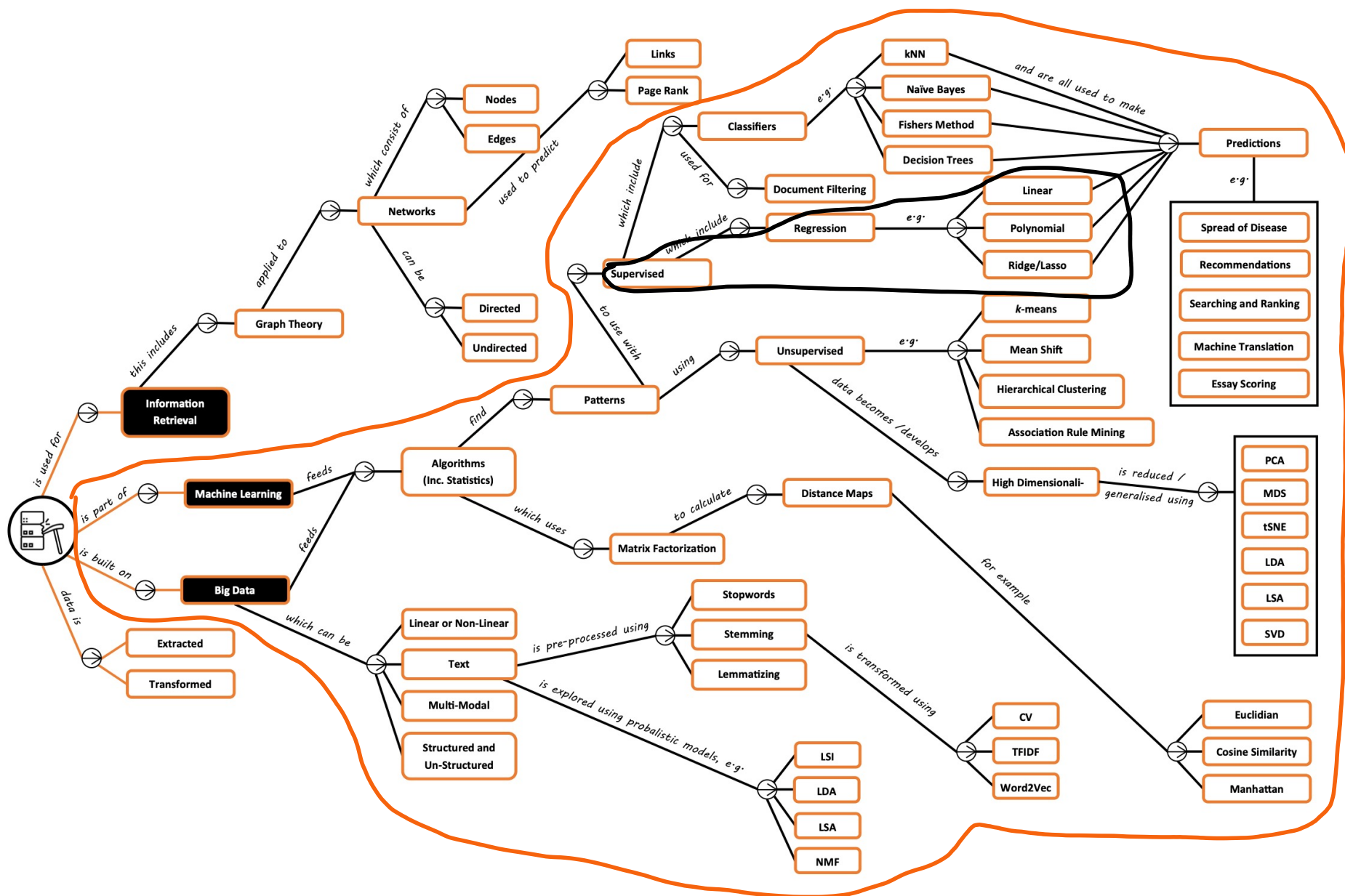
D...

	Distance to $C1=2$	Distance to $C2=8$	Assigned Center
1	$ 1-2 $	$ 1-8 $	$C1$
2	$ 2-2 $	$ 2-8 $	$C1$
3	$ 3-2 $	$ 3-8 $	$C1$
4	$ 4-2 $	$ 4-8 $	$C1$
6	$ 6-2 $	$ 6-8 $	$C2$
7	$ 7-2 $	$ 7-8 $	$C2$
8	$ 8-2 $	$ 8-8 $	$C2$
9	$ 9-2 $	$ 9-8 $	$C2$
10	$ 10-2 $	$ 10-8 $	$C2$

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 - Each MCQ contains 2 or 3 subquestions
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- How to prepare?
 - Understand concepts and ideas for easy/normal MCQs
 - Cover all the content from lecture slides except for appendix ones
 - Check out overview slides + learning-outcome slides for big pictures
 - Do some calculations on learned algorithms for hard MCQs

Roadmap (My Part)



Lecture 1 – Recommendation Systems

Content-based

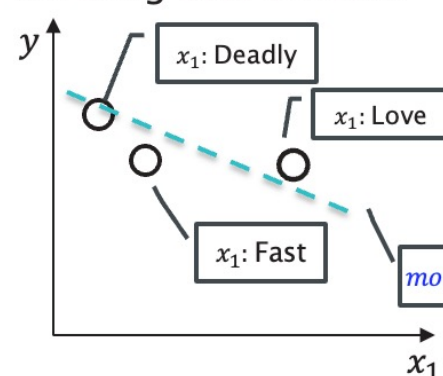
$$\min_{\theta} \frac{1}{m} \sum_{i=1}^m (\theta^T X_i - y)^2$$

$$x(\text{Love}) = (x_1, x_2) = (1, 0.1)$$

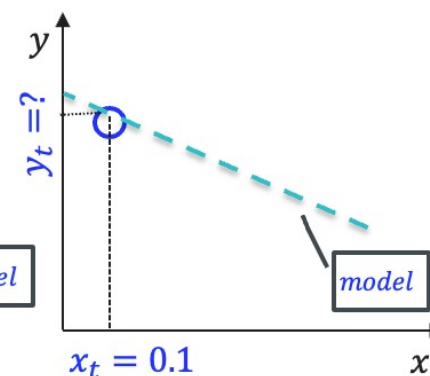
Film	Dave	x_1 romance	x_2 action
Love Really	4	1	0.1
Deadly Weapon	5	0.1	1
Fast and Cross	4	0.2	0.9
Star Battles	?	0.1	1

$y_t = ?$

1. Fitting User Behavior



2. Prediction



User-based Collaborative

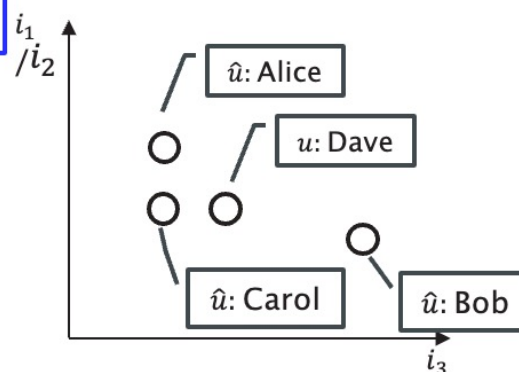
$$r_{u,i} = \frac{\sum_{\hat{u} \in U} \text{sim}(u, \hat{u}) r_{\hat{u},i}}{\sum_{\hat{u} \in U} |\text{sim}(u, \hat{u})|}$$

$$\hat{u}(\text{Alice}) = (i_1, i_3) = (4, 5)$$

	Film	Alice	Bob	Carol	Dave
i_1	Love Really	4	1		4
i_2	Deadly Weapon		1	4	5
i_3	Fast and Cross	5		5	4
	Star Battles	1	5	2	?

$r_{u,t} = ?$

1. Finding Similar Users



2. Prediction

Sim(Dave, Alice) = 0.8
Sim(Dave, Carol) = 0.6

$$r_{u,t} = \frac{0.8 \times 1 + 0.6 \times 2}{0.8 + 0.6}$$

e.g., Cosine Similarity:

$$\cos(\theta) = \frac{p \cdot q}{|p||q|}$$

$$= \frac{\sum_{i=1}^N p_i q_i}{\sqrt{\sum_{i=1}^N p_i^2} \sqrt{\sum_{i=1}^N q_i^2}}$$

Note: The similarities (0.8, 0.6) above are merely used for demonstration. With Cosine similarity, they'd be 0.99 and 0.98 if we assume the missing ratings can be replaced with the user's average on other films excluding 'Star Battles' (e.g., "Deadly Weapon" with Alice is 4.5).

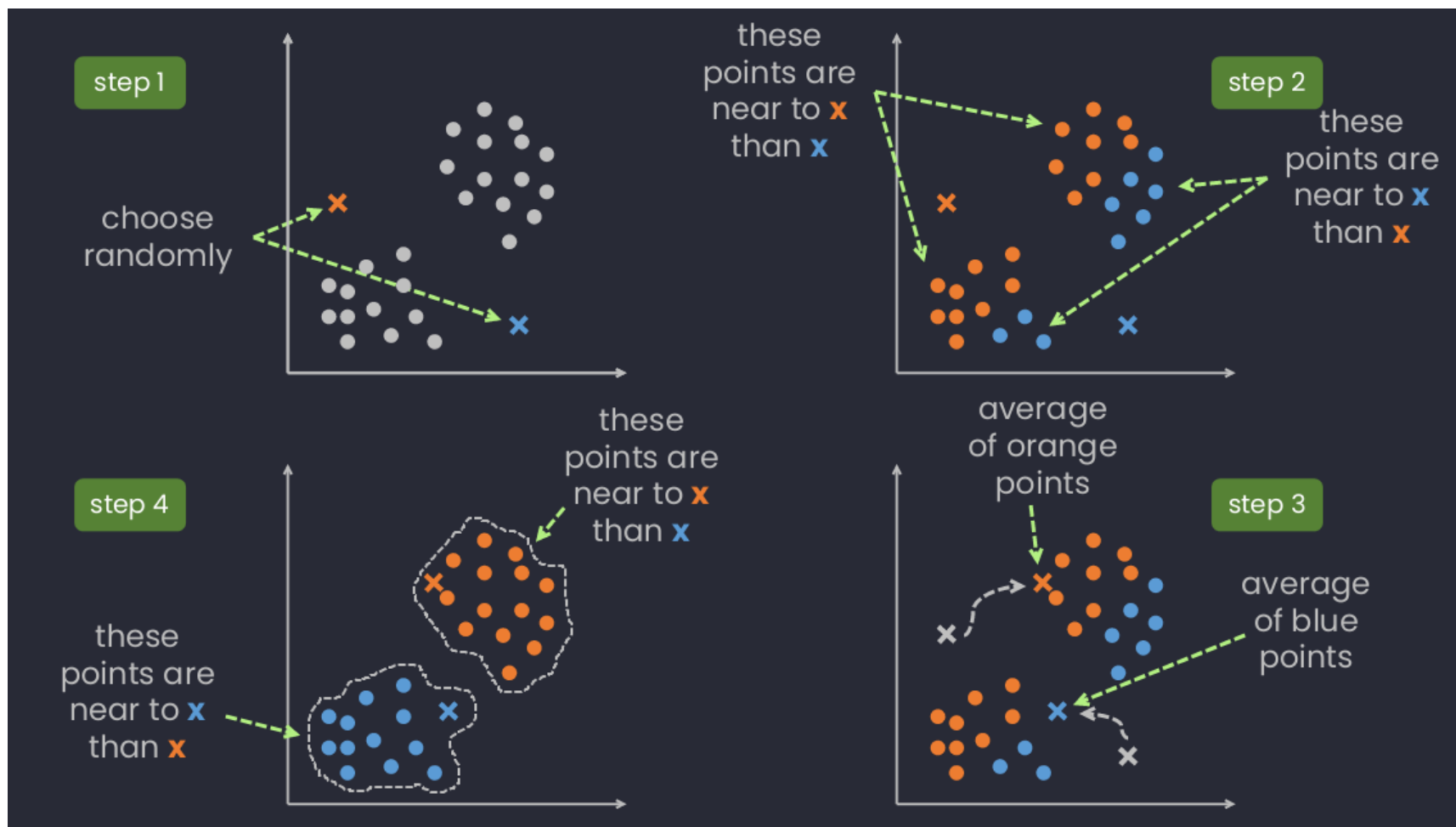
Item-based?

$$\text{sim}(i, j) = \frac{\sum_u r_{u,i} \cdot r_{u,j}}{\sqrt{\sum_u r_{u,i}^2} \cdot \sqrt{\sum_u r_{u,j}^2}}$$

- $r_{u,i}$ and $r_{u,j}$ are the ratings given by user u to items i and j .

Lecture 2 - Discovering Groups

Clustering Algorithm (1/3) – K-means



[Credit: Pratik Thorat](#)

Lecture 3 – Statistics & PCA

PCA in a nutshell

3. compute covariance matrix

$$\begin{matrix} & h & u \\ \begin{matrix} h \\ u \end{matrix} & \begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \end{matrix} \rightarrow \text{cov}(h,u) = \frac{1}{n} \sum_{i=1}^n h_i u_i$$

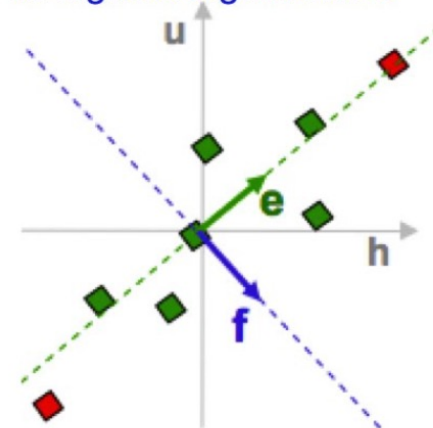
4. eigenvectors + eigenvalues

$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{bmatrix} e_h \\ e_u \end{bmatrix} = \lambda_e \begin{bmatrix} e_h \\ e_u \end{bmatrix}$$

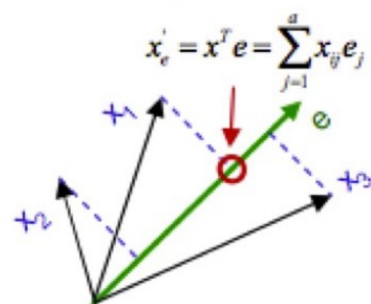
$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{bmatrix} f_h \\ f_u \end{bmatrix} = \lambda_f \begin{bmatrix} f_h \\ f_u \end{bmatrix}$$

`eig(cov(data))`

5. pick $m < d$ eigenvectors
w. highest eigenvalues



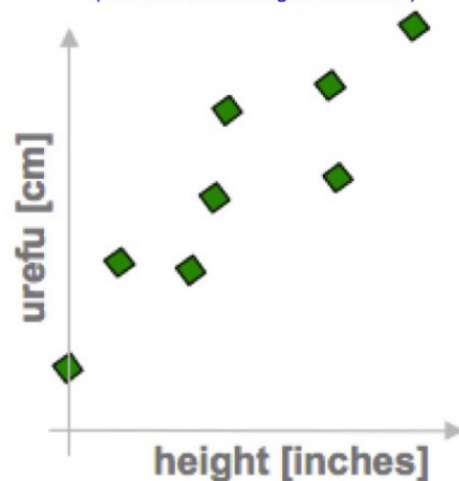
6. project data points to those eigenvectors

$$x'_e = x^T e = \sum_{j=1}^d x_{ij} e_j$$


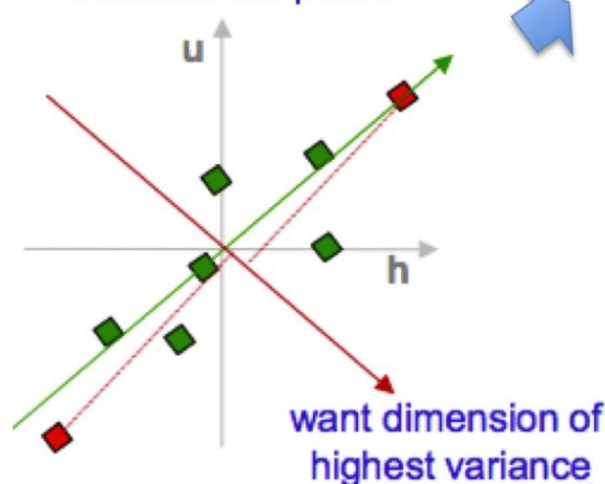
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1. correlated hi-d data

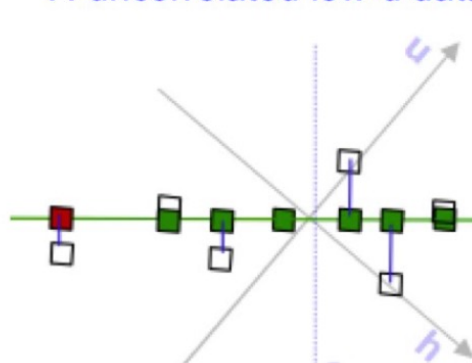
("urefu" means "height" in Swahili)



2. center the points



7. uncorrelated low-d data



Lecture 4- Embedding Data: Self-Organising Maps

To update the weight vector $\mathbf{w}$

$$\mathbf{w}(t+1) = \mathbf{w}(t) + \theta(u, v, t)\alpha(t)(x_i - \mathbf{w}(t))$$

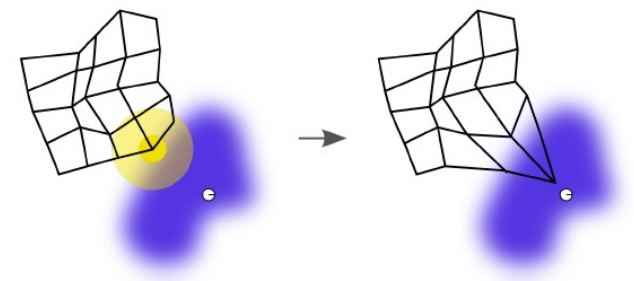
where θ is the neighbourhood weighting function (usually Gaussian)

e.g., $\theta = \begin{cases} 1 & \text{if the node is the winner} \\ 0.5 & \text{if the node is immediate neighbour of the winner} \\ 0 & \text{otherwise} \end{cases}$

α is the learning rate

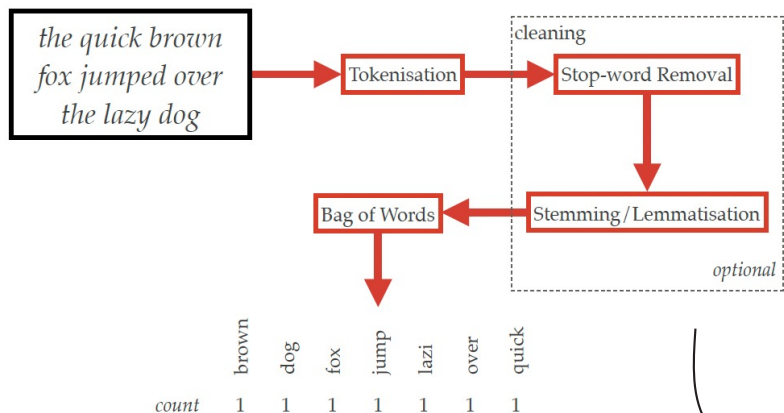
x_i is the input vector

u is the winner node, and v is the node whose weight is $\mathbf{w}$

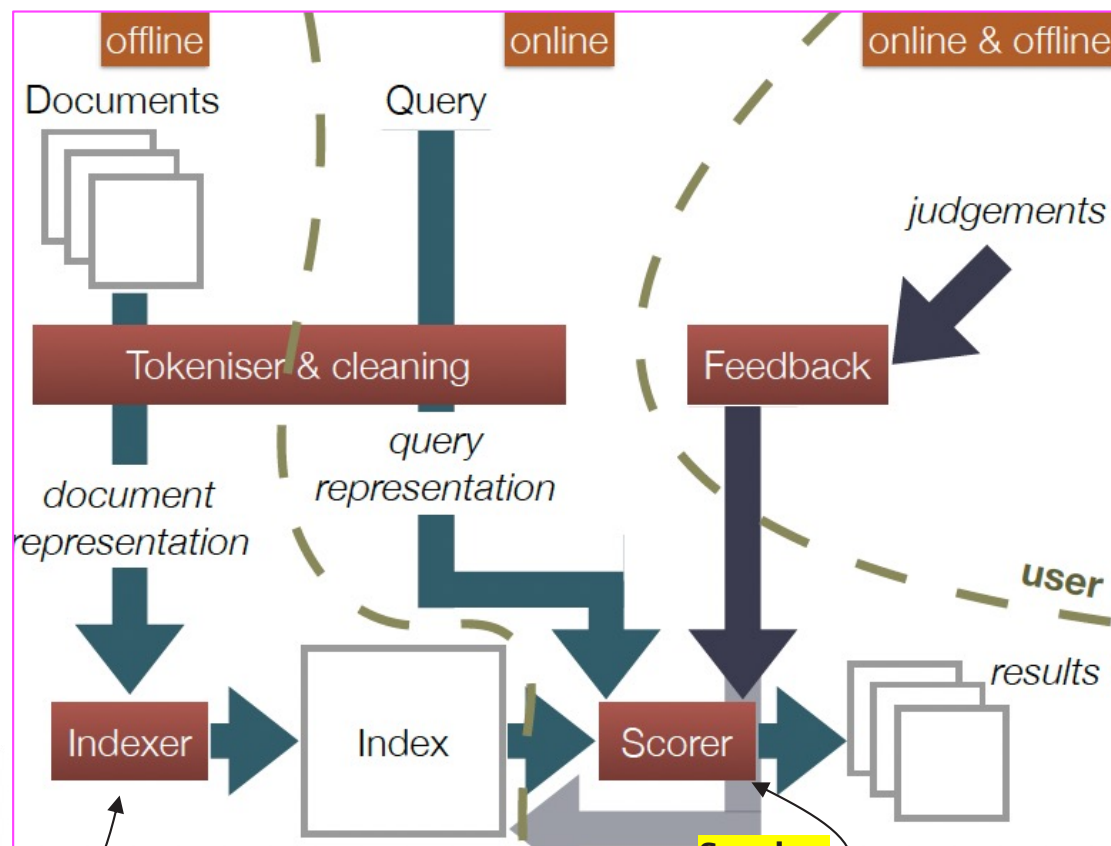
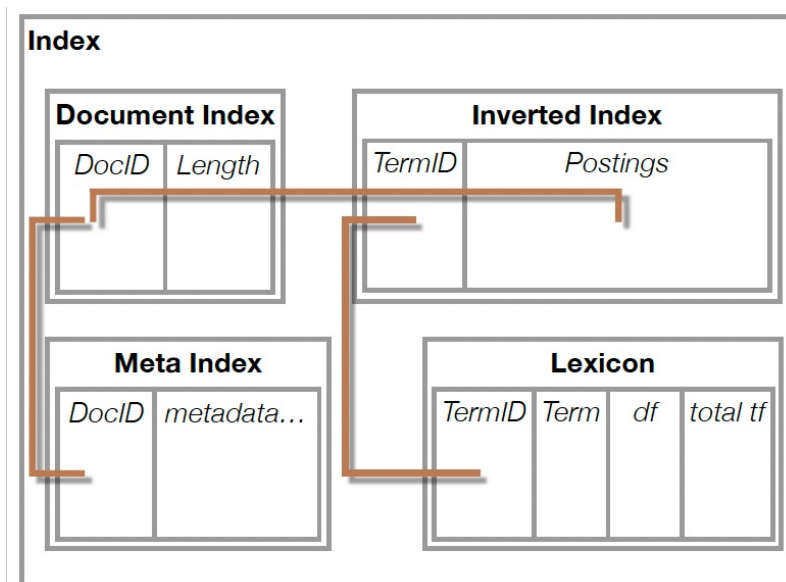


Both the learning rate and the neighbourhood weighting function get smaller over time

Lecture 5 - Search and Rank



Encoding
(vector space modeling)



Indexing
(sorting)

Scoring
(matching)

Number of times w appears in d

$$f(\mathbf{q}, \mathbf{d}) = \sum_{i=1}^N q_i y_i = \sum_{w \in q \cap d} c(w, q) c(w, d) \log \frac{M + 1}{df(w)}$$

Number of times w appears in q

document frequency
(number of docs containing w)

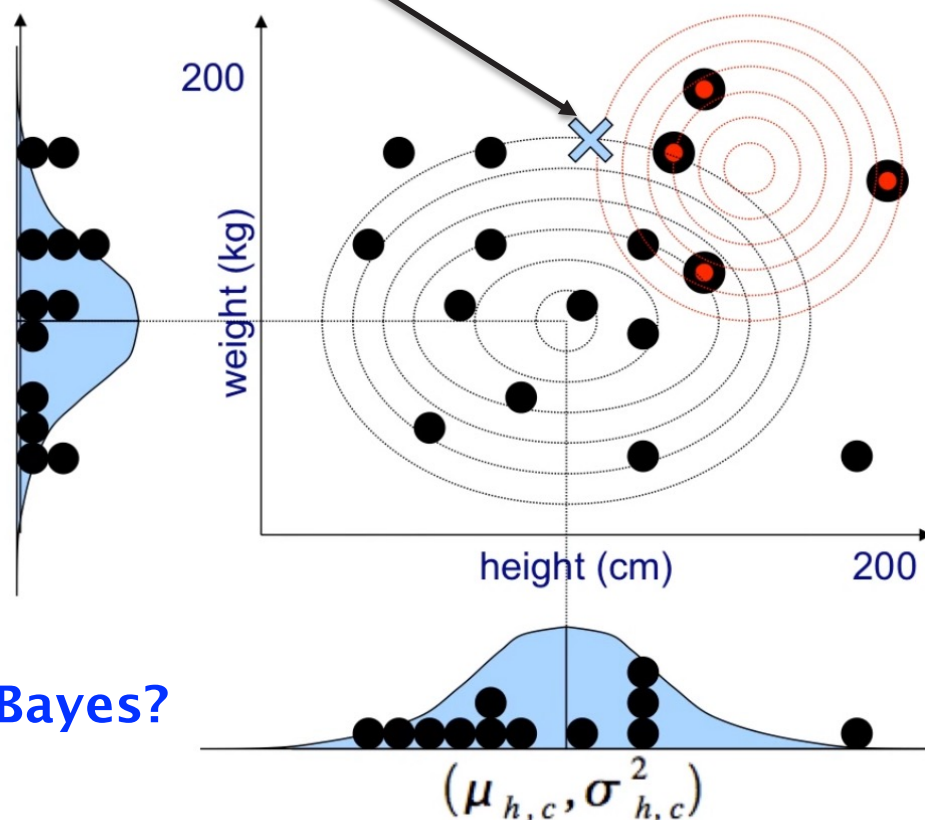
Total #docs in collection

Lecture 6 - Document Filtering



1. Priors

$$P(a) = \frac{4}{4+12} = 0.25 ; P(c) = 0.75$$



Naïve Bayes?

Fisher method? $P(a|x) = P(a|h_x)P(a|h_y)?$

2. Likelihoods

$$p(h_x|c) = \frac{1}{\sqrt{2\pi}\sigma_{h,c}} \exp - \frac{1}{2} \left(\frac{(h_x - \mu_{h,c})^2}{\sigma_{h,c}^2} \right)$$

$$p(w_x|c) = \frac{1}{\sqrt{2\pi}\sigma_{w,c}} \exp - \frac{1}{2} \left(\frac{(w_x - \mu_{w,c})^2}{\sigma_{w,c}^2} \right)$$

$$p(h_x|a) = \frac{1}{\sqrt{2\pi}\sigma_{h,a}} \exp - \frac{1}{2} \left(\frac{(h_x - \mu_{h,a})^2}{\sigma_{h,a}^2} \right)$$

$$p(w_x|a) = \frac{1}{\sqrt{2\pi}\sigma_{w,a}} \exp - \frac{1}{2} \left(\frac{(w_x - \mu_{w,a})^2}{\sigma_{w,a}^2} \right)$$

Feature independence

$$P(x|a) = p(h_x|a) p(w_x|a)$$

$$P(x|c) = p(h_x|c) p(w_x|c)$$

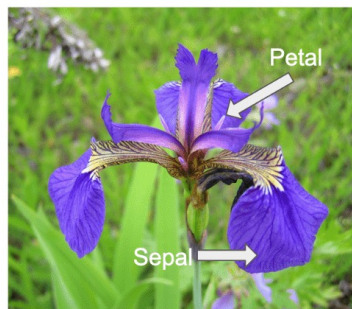
3. Posteriors

$$P(a|x) = \frac{P(x|a)P(a)}{P(x|a)P(a) + P(x|c)P(c)}$$

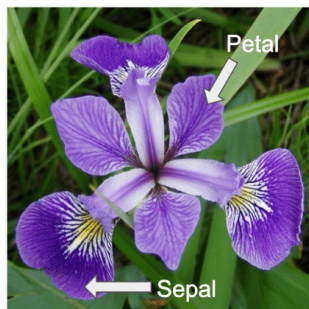
$$p(c|x) = \frac{p(x|c)p(c)}{p(x|a)p(a) + p(x|c)p(c)}$$

Lecture 7- Decision Trees

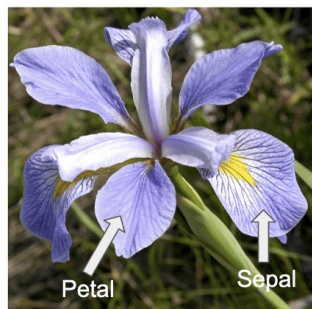
Iris setosa



Iris versicolor



Iris virginica



The objective is to efficiently partition the data based on the most informative conditions (based on features) that separate different classes.

Key idea: Greedily find best split for each feature

C0: 5
C1: 5

Non-homogeneous,
High degree of impurity

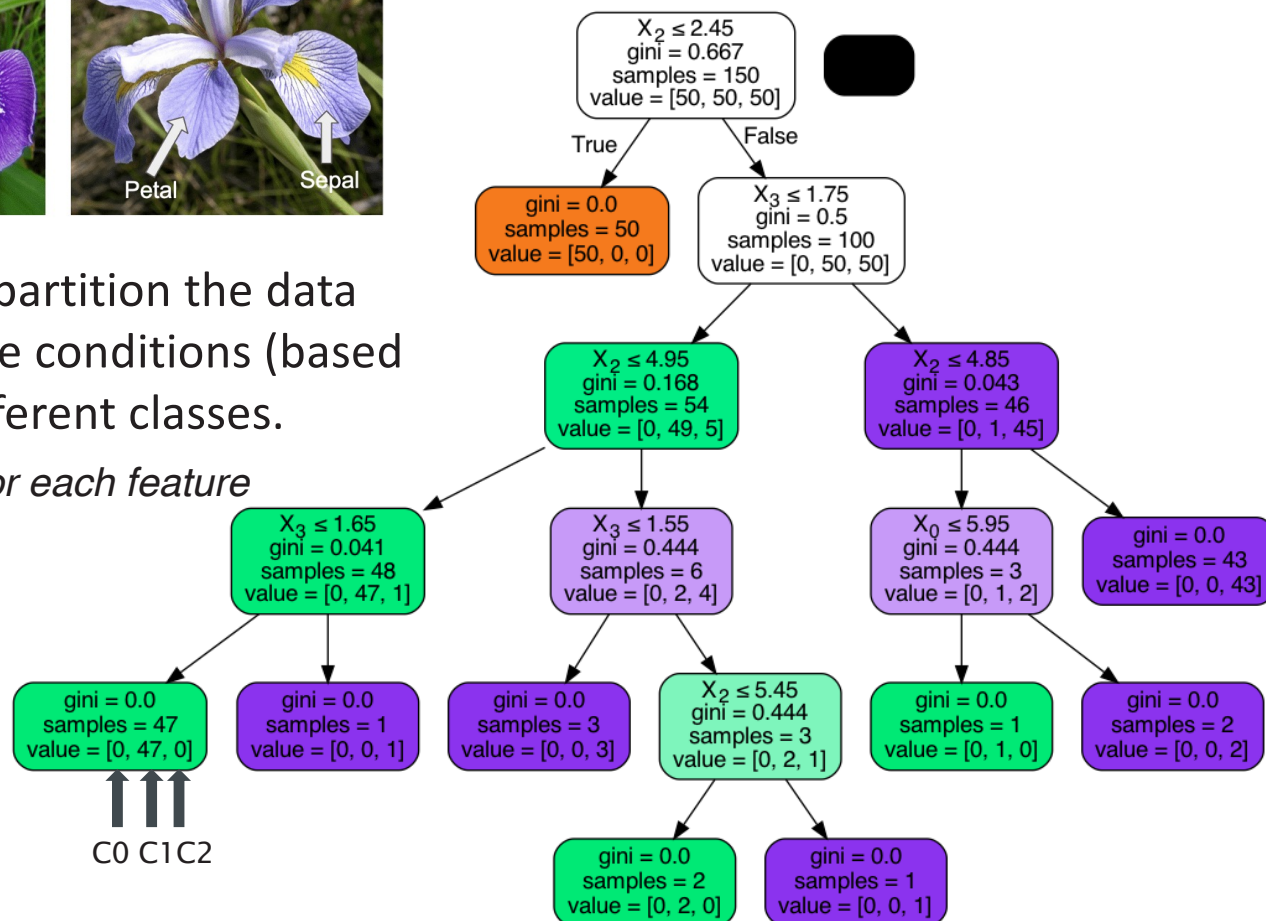
C0: 9
C1: 1



Homogeneous,

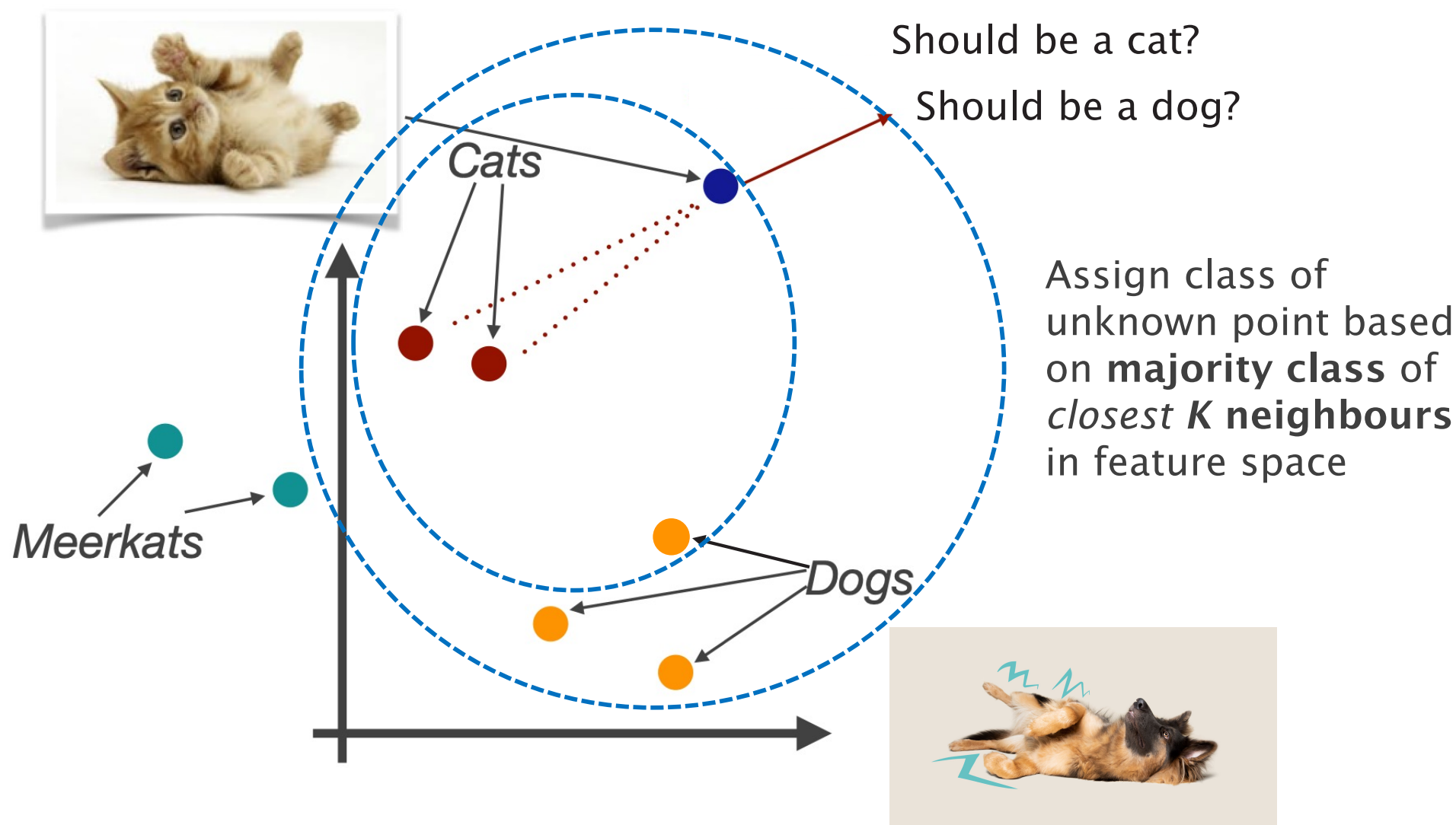
Low degree of impurity

A decision tree is like a flow chart
For example, a tree on the Iris dataset



Four features, only three are used here, and one is only used once.

Lecture 8 - Nearest Neighbours



Lecture 9- Market Basket

EXAMPLE OF ASSOCIATION RULES



Assume there are 5 customers

3 of them bought **milk**, 2 bought **potato chip** and 2 bought both of them

Transaction 1: Frozen pizza, cola, milk
 Transaction 2: Milk, potato chips
 Transaction 3: Cola, frozen pizza
 Transaction 4: Milk, potato chips
 Transaction 5: Cola, pretzels



milk → **potato chip**

support milk = $P(\text{milk}) = 3/5 = 0.6$

support potato chip = $P(\text{potato chip}) = 2/5 = 0.4$

support = $P(\text{milk \& potato chip}) = 2/5 = 0.4$

confidence

= support (milk & potato chip) / support(milk)

= $0.4/0.6$

= 0.67

CONFIDENCE = $P(\text{Milk \& potato chip}) / P(\text{Milk})$



How about
 Potato chip
 → Milk ?

Diagram illustrating the matrix factorization process:

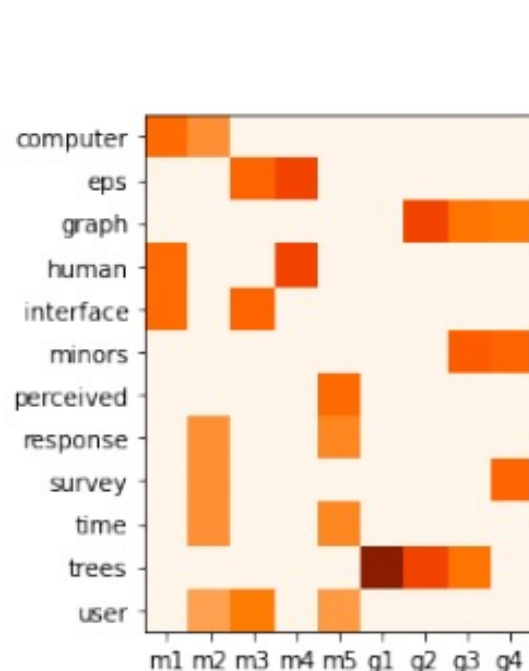
- Matrix A (dimensions $m \times n$) is labeled "docs" (vertical) and "words" (horizontal).
- Matrix U_r (dimensions $m \times r$) is labeled "words" (vertical) and "m x r" (horizontal).
- Matrix Σ_r (dimensions $r \times r$) is labeled "concepts" (vertical) and "r x r" (horizontal).
- The approximation is shown as $A \approx U_r \Sigma_r$.

- ▶ This means it is proportional to the covariance or correlation between the documents
- ▶ These are the *concepts*

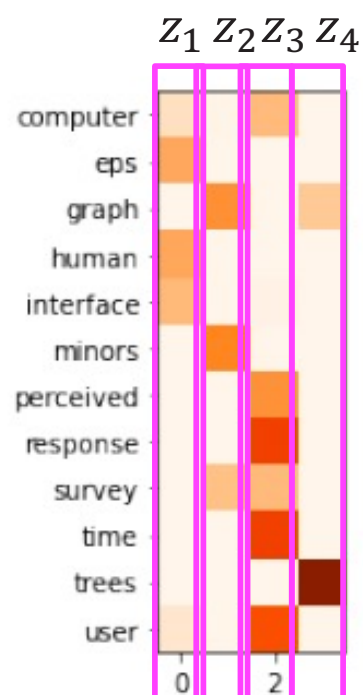
17/34

Lecture 11 - Topic Modelling

Non-negative Matrix Factorisation (NMF)

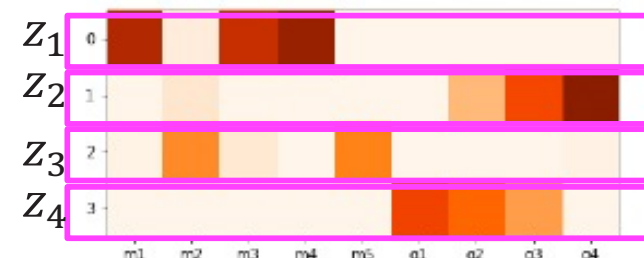


A



W

z_i : i -th topic



H

W has the *basis vectors*, showing how the words are clustered
 H has the topic memberships for the documents.

Lecture 12 - Outlier Detection

